

Group 37 MATH 189 Project Proposal

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Background

San Diego's rental market is among the most unaffordable in the United States. A [2024 study by researchers at Chapman University](#) names San Diego as an "impossibly unaffordable" city and the 4rd most unaffordable in America, ranking just below the notoriously expensive Silicon Valley hubs of San Francisco and San Jose. A widely-debated cause of this crisis is the increase of investment in short-term rental units, eating away at the housing supply in popular tourist neighborhoods such as La Jolla and Pacific Beach. There is strong evidence and research supporting this concern. A [study of Barcelona's housing market](#) showed that Airbnb raised average neighborhood rents by nearly 2%, with the increase rising to as much as 7% in neighborhoods with the highest density of Airbnbs. A [similar study](#) conducted in Berlin found that "each additional commercial Airbnb listing displaces 0.23 to 0.37 rental units and increases rent per square meter by 1.3 to 2.4 percent."

In response to the crisis, [San Diego implemented the Short-Term Residential Occupancy \(STRO\) ordinance in May 2023](#), which capped whole-home short-term rental units at 1% of the city's housing stock. This recent policy shock presents a solid opportunity for us to analyze whether the city's attempt to combat housing affordability concerns was successful.

Question

Did the city of San Diego's May 2023 STRO ordinance, which eliminated nearly 50% of short-term rentals across the city, cause a statistically significant decrease in long-term rent prices, and was this effect stronger in ZIP codes with higher pre-ordinance Airbnb listing density?

Hypothesis

Null Hypothesis: The implementation of short-term rental caps **did not** have a differential effect on the average Airbnb listing price between both high-density and low-density areas.

Alternative Hypothesis: The implementation of short-term rental caps **did** have a differential effect on the average Airbnb listing price between both high-density and low-density areas.

Data

The dataset we are going to be using is publicly available consisting of Airbnb listing data for properties located in San Diego. We will be directly downloading the CSV files provided by insideairbnb.com. Each observation represents an individual listing, the key features that we will be analyzing are price, neighborhood, listing date, latitude, and longitude. Using these features, we will classify each listing into either high or low density areas. We will then compare average prices before and after the implementation of the short-term rental ordinance.

Analysis Methods

To evaluate the effect of the May 2023 short-term rental ordinance on Airbnb listing prices, we will apply multiple linear regression and hypothesis testing techniques. Our analysis will follow consist of four different aspects:

1. **Exploratory Data Analysis (EDA):** We will compare Airbnb price distributions in high- and low-density neighborhoods using Quantile-Quantile plots and empirical distribution functions to assess variance, skewness, and baseline trends.

2. **Multiple Linear Regression Modeling:** Using ordinary least squares, we will model the ordinance's differential impact using Density and Time as categorical predictors, focusing on their interaction.
3. **Regression Results:** We will evaluate model fit using R^2 and verify assumptions of linearity, normality, and homoscedasticity via residual plots and ACF. Cook's distance and leverage metrics will be used to handle influential outliers.
4. **Hypothesis Testing:** We will test the interaction term's significance to determine if the ordinance's effect varies by density. If regression assumptions are irreparably violated, we will use the nonparametric Two-Sample Kolmogorov-Smirnov test to compare price changes.

Expected Results

Based on basic economic principles of supply and demand, we expect that the strict enforcement of the short-term rental ordinance significantly reduced the supply of short-term rentals in high-density areas, while leaving low-density areas relatively unaffected. Therefore, we anticipate the following outcomes:

- A statistically significant interaction term ($p < 0.05$), confirming the ordinance had a differential effect based on neighborhood density.
- Right-skewed price distributions with heavy tails, likely requiring data transformations or outlier mitigation to satisfy regression assumptions.
- A disproportionate increase in average listing prices in high-density neighborhoods compared to the natural market fluctuations in the low-density control group.

Objective / Goals of Project

Our project goal is to see if San Diego's May 2023 short-term rental ordinance affected Airbnb prices differently in high-density compared to low-density Airbnb areas. We expect the ordinance to have a stronger effect in areas with many Airbnbs because those neighborhoods were more directly affected by the short-term rental limits. We will first use Airbnb listing data to group neighborhoods or ZIP codes by density. Then, we will compare average Airbnb prices before and after the ordinance to check which group's prices changed more. To test this hypothesis, we will use the methods listed above. If high-density Airbnb areas show a larger price change after the ordinance, it would suggest that the policy had a stronger impact in those neighborhoods. If the changes are similar between the two groups, then the ordinance may not have had a clear differential effect on Airbnb prices.

Timeline

Week 7: Data Collection and Cleaning

We will download the Airbnb listing data and clean the main variables, such as price, neighborhood, latitude, longitude, and date. We will also split the data into pre- and post-May 2023 ordinance periods.

Week 8: Density Groups and Exploratory Data Analysis

We will classify neighborhoods into high-density and low-density groups. Then we will create plots such as histograms, box plots, QQ plots, and empirical distribution functions to compare prices across groups and identify outliers.

Week 9: Regression and Hypothesis Testing

We will run our main regression model with Airbnb prices as the response variable. The key part of the model will be the interaction between the density group and the time period. We will use this to test whether the ordinance affected high-density areas differently from low-density areas.

Week 10: Final Report and Video Presentation

We will finish the final report, organize our plots and results, and explain our conclusions. We will also make slides and record the video presentation. We will use this week to finalize and submit everything.